SAT Hard Question 16

1 ★ Mark for Review **ABC**

The function h is defined as:

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant.

The function estimates that the object is 40.96 feet above the ground when it is dropped at $t = 0$.

How many seconds after being dropped does the function estimate the object will hit the ground?

2 ★ Mark for Review **ABC**

For a set of three consecutive integers, the result of subtracting the greatest integer from the sum of the other two integers is 48.

What is the value of the greatest of the three integers?

3 ★ Mark for Review **ABC**

A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was $7,150\%$ of the number of postdoctoral students surveyed, and the number of graduate students surveyed was 45% of the number of undergraduate students surveyed.

If there were $6,435$ graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

4 ★ Mark for Review **ABC**

$$16x - 20y = 6y + 8$$

$$ty = 8x + \frac{1}{5}$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

5 ★ Mark for Review ABC

The bone mineral density of a certain person's thoracic spine was estimated to be 0.812 grams per square centimeter. What is this estimated bone mineral density in grams per square millimeter? (1 centimeter = 10 millimeters)

A. 81.2

B. 8.12

C. 0.0812

D. 0.00812

6 ★ Mark for Review ABC

$$\frac{5}{9}(5x+9)(x+\sqrt{5k+9})(x-\sqrt{5k+9})=0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 72. What is the value of k?

7 ★ Mark for Review ABC

A right rectangular prism has a base area of $56t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{14}{3}$ cm, and the height of the rectangular prism is 3 cm.

Which expression represents the surface area, in cm^2, of the right rectangular prism?

A. $112t+56$

B. $184t+28$

C. $1,680t+28$

D. $168t$

8 ★ Mark for Review ABC

In the xy-plane, a unit circle with center at the origin O contains point A with coordinates $(1, 0)$ and point B with coordinates $\left(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}}\right)$.

If the measure of angle AOB is p radians, what is the value of $\frac{\cos p}{\sin p}$?

9 ★ Mark for Review ABC

The function h is defined by:

$$h(x) = -\sqrt{x^2 + bx + c}$$

where b and c are constants. In the xy-plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $\left(0, -\sqrt{334}\right)$.

If $h(m) = 0$, what is the greatest possible value of m?

10 ★ Mark for Review ABC

A computer repair specialist charges \$160 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 6 hours of repair is \$360.

Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 50x + 60$

B. $f(x) = 50x + 160$

C. $f(x) = 60x$

D. $f(x) = 60x + 160$

11 ★ Mark for Review ABC

A painting studio is offering a promotion where the first class is free, the second class is half off the regular price, and the remaining classes are regularly priced. If the regular price of a class is \$26.40, which function f gives the total cost, in dollars, of x classes taken using this promotion, where $x > 2$?

A. $f(x) = 26.40(x - 1) + 13.20$

B. $f(x) = 26.40(x - 2) + 13.20$

C. $f(x) = 26.40(x - 1) + 13.20(x - 2)$

D. $f(x) = 26.40(x - 2) + 13.20(x - 1)$

12 ★ Mark for Review ABC

The function f is defined by the given equation:

$$f(x) = x^2 - 4x - 320$$

Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

A. $f(x) = x^2 - 4(x + 80)$

B. $f(x) = (x - 2)^2 + (-324)$

C. $f(x) = x(x - 4) + (-320)$

D. $f(x) = (x + 16)(x - 20)$

13 ★ Mark for Review ABC

The composition of an animal is defined as the muscles, bones, and fat of the animal. A scientist studied the composition of one young swamp buffalo and determined the buffalo had 133 kilograms of muscle, which made up approximately 65.9% of its composition. Of the remaining composition of this buffalo, approximately 46.2% was bone, and the remainder was fat. Based on these approximations, to the nearest tenth, how many kilograms of this buffalo's composition was bone?